

02.06 Surveys, Experiments, and Observational Studies

Practice problems

Show your work and state restrictions or excluded values when needed.

Recognize	Apply these rules
Statistics And Probability	Identify the population, variable, or probability model before calculating.
The expression contains the chapter's key notation or model.	Write the known quantities, choose the matching formula, calculate carefully, and check the result in context.

A. Practice: Surveys, Experiments, and Observational Studies

1.

$$P(A) = \frac{3}{10}$$

2.

$$\mu = \frac{12 + 15 + 18 + 20}{4}$$

3.

$$Z = \frac{82 - 70}{8}$$

4.

$$P(A) = \frac{3}{10}$$

5.

$$\mu = \frac{12 + 15 + 18 + 20}{4}$$

6.

$$Z = \frac{82 - 70}{8}$$

7.

$$P(A) = \frac{3}{10}$$

8.

$$\mu = \frac{12 + 15 + 18 + 20}{4}$$

9.

$$Z = \frac{82 - 70}{8}$$

10.

$$P(A) = \frac{3}{10}$$

11.

$$\mu = \frac{12 + 15 + 18 + 20}{4}$$

12.

$$Z = \frac{82 - 70}{8}$$

13.

$$P(A) = \frac{3}{10}$$

14.

$$\mu = \frac{12 + 15 + 18 + 20}{4}$$

15.

$$Z = \frac{82 - 70}{8}$$

16.

$$P(A) = \frac{3}{10}$$

17.

$$\mu = \frac{12 + 15 + 18 + 20}{4}$$

18.

$$z = \frac{82 - 70}{8}$$

19.

$$P(A) = \frac{3}{10}$$

20.

$$\mu = \frac{12 + 15 + 18 + 20}{4}$$

B. Apply the key formulas

1.

$$P(A) = \frac{\text{favorable}}{\text{total}}$$

2.

$$\mu = \frac{\sum b}{n}$$

3.

$$Z = \frac{c - \mu}{\sigma}$$

4.

$$P(A) = \frac{\text{favorable}}{\text{total}}$$

5.

$$\mu = \frac{\sum e}{n}$$

6.

$$Z = \frac{a - \mu}{\sigma}$$

7.

$$P(A) = \frac{\text{favorable}}{\text{total}}$$

8.

$$\mu = \frac{\sum c}{n}$$

9.

$$Z = \frac{d - \mu}{\sigma}$$

10.

$$P(A) = \frac{\text{favorable}}{\text{total}}$$

11.

$$\mu = \frac{\sum a}{n}$$

12.

$$Z = \frac{b - \mu}{\sigma}$$

13.

$$P(A) = \frac{\text{favorable}}{\text{total}}$$

14.

$$\mu = \frac{\sum d}{n}$$

15.

$$Z = \frac{e - \mu}{\sigma}$$

16.

$$P(A) = \frac{\text{favorable}}{\text{total}}$$

17.

$$\mu = \frac{\sum b}{n}$$

18.

$$Z = \frac{c - \mu}{\sigma}$$

19.

$$P(A) = \frac{\text{favorable}}{\text{total}}$$

20.

$$\mu = \frac{\sum e}{n}$$